

resides on untrusted BYOD devices, along with a mix of personal and professional data.

Another challenge is that the enterprise data to be accessed from mobile devices usually includes a variety of data sources at the backend, which may involve integrating standard data sources such as Salesforce, SAP and others. Or even integration of custom data sources.

Further, integration with the current enterprise ecosystem, including cloud services leveraged for computing or storage purposes as well as third party solutions such as enterprise mobility management, is crucial for making the solution work smoothly in the current set-up.

Another challenge is to build and test solutions for an ever-evolving and dense device portfolio with a variety of device platforms, device models and form factors to be targeted.

In spite of strategising and huge efforts invested in building an efficient well-integrated mobile solution, ensuring user adoption and acceptance for it is no easy task. User experience, including usability, is a vital factor impacting adoption, specially for mobility solutions.

Further, life cycle management of the solution is a major challenge considering the scale of target platforms, devices and apps, with the further complexity of supporting multiple OS versions together with multiple app versions.

One more key challenge is policy management, reporting and administration. The organisations need to be able to report enterprise-wide on everything users have done with valuable content, and integrate personal devices effectively into existing enterprise compliance solutions. They also need the ability to simply provision and on-board users and manage their environments to suit their business groups' needs and policies.

Finally, the enterprise mobile solution has to be dynamic enough to analyse, learn and evolve with time to better suit the users' needs.

So, is there a simple and quick solution to these challenges? The answer is yes.

The solution – MADP and EMM

There are two types of solutions that work together to address all the challenges to building a secure and manageable enterprise mobile solution. These are mobile application development platforms, a.k.a. MADP, and enterprise mobility management solutions or EMM.

MADP empowers cross-platform mobile app development for enterprise apps to address platform diversity and ensure that they integrate well with the existing enterprise ecosystem. It helps to get modular access to heterogeneous data sources, and all this along with effective handling of the app life cycle.

EMM focuses on securing data at rest and data on the move for enterprises. It also provides remote manageability for device, app and content. It has analytics provision too, thereby helping in identifying and addressing low user adoption.

Let's take a closer look at both these solutions.



Figure 1: Gartner's Magic Quadrant for MADPs, July 2015

Mobile Application Development Platforms (MADP)

According to Gartner, "A MADP enables an enterprise to design, develop, test, deploy, distribute, analyse and manage a portfolio of cross-platform mobile apps on a range of devices running Android and iOS, and to address the requirements of diverse use cases including external-facing and internal-facing scenarios."

There are generic MADPs available in the market as products or offered by service providers in the form of white-labelled solutions. Whether to use a generic MADP or opt for a custom one largely depends on the complexity of the enterprise's ecosystem, its time-to-market needs, security concerns, and customisation requirements.

The MADP advantage

MADP brings in excellent support for cross-platform mobile app development. This enables the developer to code once and generate multiple apps targeting different platforms as per business needs.

Next, MADP helps to solve the complexity of integrating heterogeneous data sources by providing out-of-the-box integration with standard data sources like Siebel, Salesforce, Postgres, etc. It also supports custom components to integrate with legacy and custom data sources.

Further, MADPs support standard cloud services and third party tools like enterprise mobility management (EMM) solutions. Enterprises may opt for standard or custom solutions for these too.

Finally, these platforms can enable employees to build, test, distribute, manage and update apps more rapidly.

Definitely, if you're going to invest in enterprise mobile solutions, then you need to seriously consider investing in a MADP solution, whether standard or custom.

You can see Gartner's Magic Quadrant for MADPs, July 2015, in Figure 1.

It is evident that the MADP market continues to mature,